NetFix

Explainable Home and Last-Mile Connectivity Diagnosis

PROJECT OVERVIEW

- NetFix is a web-based application designed to help users understand internet connectivity problems.
- It analyzes network information such as latency, packet loss, and download speed.
- It provides understandable explanations of possible connectivity issues.
- It suggests troubleshooting steps to help users resolve common problems.
- It presents the diagnosis through a simple, interactive dashboard
- **Main Goal:** To make internet troubleshooting easier, clearer, and more accessible to everyone.

INNOVATION

- **What Makes NetFix Different?**
- **Explainable Diagnosis:** Explains possible network problems instead of showing only technical measurements.
- **Evidence-Based Analysis:** Uses connectivity indicators to identify potential issues.
- **Guided Troubleshooting:** Provides step-by-step suggestions based on the diagnosis.
- **Connection Path Visualization:** Displays the device, home router, ISP connection, public internet, and DNS as separate stages.
- **Continuous Monitoring:** Allows users to repeat scans and observe changes in latency.
- **Interactive Testing:** Includes sample scenarios such as router failure, ISP outage, DNS failure, and weak Wi-Fi.
- **Key Innovation:** Combining network measurements, understandable explanations, and troubleshooting guidance in one interface.

ARCHITECTURE

System Architecture

- The application follows a browser-based architecture with four main components.

1. User Interface

- Built using HTML and CSS.
- Collects user inputs and displays results.
- Provides interactive controls and sample scenarios.

2. Data Collection

- Uses JavaScript and browser APIs.
- Measures the response time of browser requests to an external endpoint.
- Accepts manually entered network measurements.

3. Analysis and Diagnosis

- Applies rule-based logic to the available measurements and connectivity inputs.
- Identifies possible problems based on predefined conditions.
- Generates explanations and confidence indicators.

4. Results and Recommendations

- Displays the diagnosis and supporting evidence.
- Suggests troubleshooting steps.
- Maintains recent diagnosis history in browser storage.
- Generates a downloadable text report.

Architecture Flow:

- User → Web Interface → Data Collection → Diagnosis Engine → Recommendations → Dashboard

IMPLEMENTATION AND TECHNOLOGIES

- **How We Developed NetFix**
- **HTML**
- Creates the webpage structure.
- Organizes input fields, buttons, dashboard sections, and history.
- **CSS**
- Provides responsive layouts for mobile and desktop.
- Implements colors, status indicators, cards, and visual elements.
- Supports light and dark themes.

- **JavaScript**
- Collects and processes user inputs.
- Implements rule-based connectivity diagnosis.
- Displays explanations and recommendations.
- Runs repeated scans and updates the latency graph.
- Stores recent diagnoses using local storage.
- Generates downloadable reports.
- **Key Modules**

1. Live Scan
2. Manual Diagnosis
3. Sample Scenarios
4. Latency Monitoring
5. Diagnosis History
6. Report Generation

OUTCOME

- **Expected Outcomes and Current Prototype Capabilities**
- A functional web interface for connectivity diagnosis.
- Interactive testing using predefined network scenarios.
- Identification of potential connectivity issues using rule-based logic.
- Understandable explanations for the generated diagnosis.
- Troubleshooting suggestions for common problems.
- A visual connection-path indicator.
- Repeated browser-request timing measurements displayed as a graph.
- Browser-based diagnosis history and downloadable reports.
- **Result:** NetFix demonstrates how network measurements and rule-based analysis can be combined to make connectivity troubleshooting easier to understand.

TESTING AND VALIDATION

Testing the NetFix Prototype
The following scenarios can be used to evaluate the application.

Test Scenario	Expected Diagnosis
Healthy connection	No connectivity fault found
Router unreachable	Possible home-network issue
Internet unreachable	Possible ISP or last-mile issue
DNS failure	Possible DNS resolution issue
High packet loss	Unstable connection
High latency	Slow or delayed connection
Low download speed	Low throughput

- **Validation Approach**
- Check whether each scenario produces the expected diagnosis.
- Verify that the correct evidence and recommendations are displayed.
- Test the interface on mobile and desktop screens.
- Check monitoring, history, and report-download functions.
- *These are proposed validation cases. Include actual test results only after running the tests.*

LIMITATIONS AND FUTURE SCOPE

Current Limitations

- Browser restrictions prevent direct access to certain network diagnostics.
- The current live scan does not perform a true ICMP ping or direct packet-loss test.
- Router reachability and DNS status are entered by the user rather than independently verified.
- The rule-based system identifies possible causes but cannot always determine the exact fault.

Future Enhancements

- Integrate reliable network measurement services or a backend diagnostic agent.
- Add direct router, gateway, DNS, and packet-loss checks where supported.
- Improve diagnosis using more network measurements and historical trends.
- Introduce machine-learning-based fault classification after collecting suitable data.
- Add downloadable PDF reports and visual performance summaries.

CONCLUSION

- NetFix addresses the difficulty of understanding common internet connectivity problems.
- It combines network indicators, rule-based analysis, and troubleshooting recommendations.
- Its interactive dashboard helps users interpret potential network faults.
- The current prototype provides a foundation for developing a more comprehensive connectivity diagnosis system
- **Final Message:** NetFix aims to turn confusing network measurements into clear explanations and practical troubleshooting steps.